

Term Three Holiday Work

1a) 4 4 5 5 5 6 $\Sigma x = 29$
 the number must be "6" so that
 median is 5 and mean is 5

1b) the number must still be 6.

4 4 5 5 5 6 6

$\therefore$ mode is 5 and mean is 5

2a) mean : Size 11
 median : Size 11
 mode : Size 11

$$\frac{99910101010111111111111112131414}{20} = 11$$

2b) For an owner of a shoe store, I would assume the mode is the most applicable so that he orders more pairs of these shoes.

$$4) \frac{(56 \times 4) + (44 \times 6)}{10} = 48.8\%$$

$\therefore$ This student will have to repeat

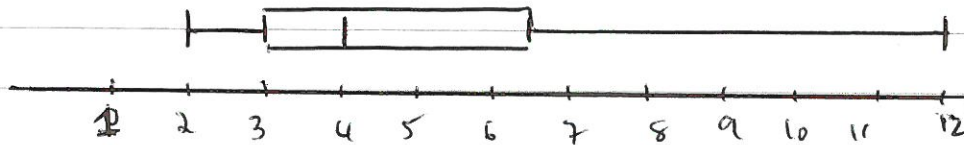
5a) mean = \$52791.67
 mode = \$31000
 median = \$32750

5b) They would have used the "median"

5c) They would have used the "mean"

6a) CN 6b) CN 6c) NC 6d) CN 6e) ND 6f) ⁶⁰

7a)



7b) Skewed to the right
Median - Q_3 is spread 3 Q_3 - max is very spread.

$$\begin{aligned} 7c) \quad Q_1 - 1.5IQR &\leq \text{Scores} \leq Q_3 + 1.5IQR \\ 3 - 1.5(3.5) &\leq \text{Scores} \leq 6.5 + (1.5 \times 3.5) \\ -2.25 &\leq \text{Scores} \leq 11.75 \end{aligned}$$

$\therefore$ Yes an outlier is present (12)

7d) Removing the 12 will change the max score to 8 instead of 12.

8) Class A's median (29), upper & lower quartiles (36 & 27) are higher than that of class B. Overall, Class A performed better in this test than class B.

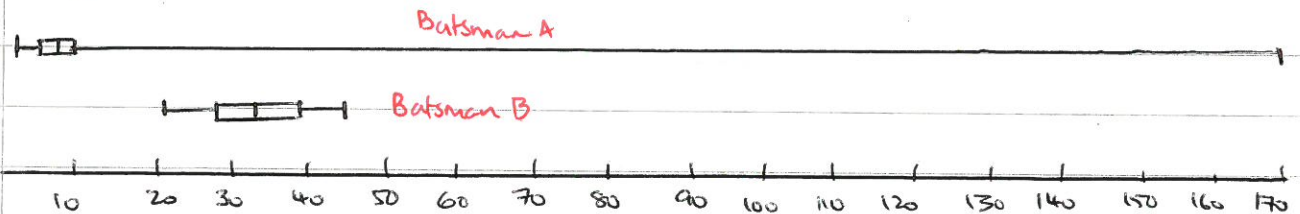
9a) 25% of his colleagues scored better than him
 $\therefore 0.25 \times 180 = 45$ colleagues scored better than him.

9b) This statement is invalid as these sections (min - Q_1) and (Q_3 - max) have the same amount of data. Q_3 - max is more spread out.

10a) Numerical discrete

10b) A: 34 runs B: 32.5 runs

10c)



10d) Batsman A should lose his spot. Batsman A has a higher mean, his median is significantly lower than B. The boxplot of Batsman A is skewed with a large range, indicating he is an inconsistent scorer.

11a) $\bar{x} = 6.06 \text{ mm}$

11b) The original data has been grouped

11c) The data is negatively skewed, unimodal with the modal class being 6.5 - 7.5 mm. The range is 9 mm.

12a) Stem and leaf "A"

12b) Stem and leaf "B"

13ai) The mean will increase

13aiv) The standard deviation will increase

14a) Japanese: mean is 62.25%

14b) Quinn was more consistent in Italian as his standard deviation is significantly lower ($8.07 < 17.46$)

15a) Machine 1

mean: 209.92

std. dev: 3.05

Machine 2

mean: 210.84

std. dev: 3.49

15bi) Machine 1 as it has
a mean closer to 210

15bii) Machine 2 as it has
the smaller standard
deviation

16a) 68%.

16b) 95%.

16c) 16%.

17) 2.5%.

18a) 84%.

18b) 81.5%.

18c) 68%.

19) Standard deviation is 4

20) 0.3944

21a) 0.5249

21b) 151.80mL

22) In comparison to the mean of each subject, Julia
attained a better result in French, achieving a score
1.5 standard deviations above the mean

$$23a) 0.8944 = 89.44\%$$

$$23b) 0.7887 = 78.87\%$$

$$23c) 8.84 \times 10^{-5} \times 100\,000$$

$$= 8.84$$

= 9 patients

24a) Bearing of A from B is 200° .

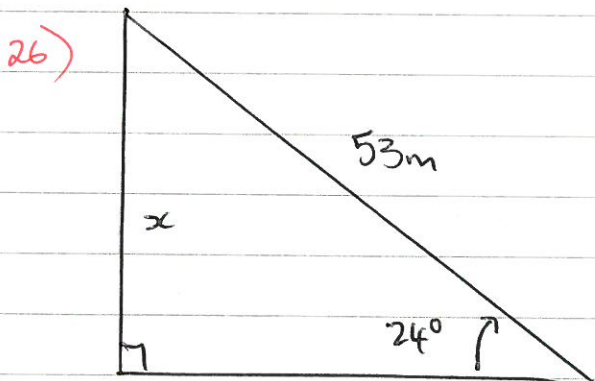
24b) Bearing of A from B is 290° .

$$25) \sin 30 = \frac{8}{x}$$

$$x = \frac{8}{\sin 30}$$

$$x = \frac{8}{0.5}$$

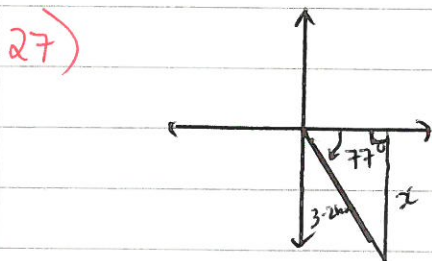
$$x = 16 \text{ cm}$$



$$\sin 24 = \frac{x}{53}$$

$$\therefore x = 21.56 \text{ m}$$

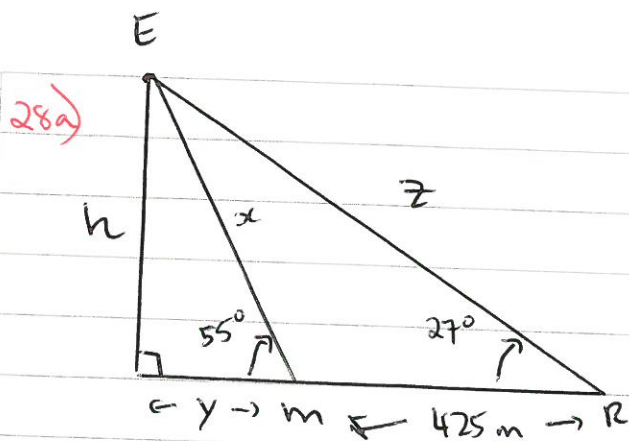
$\therefore$ Pole is 21.56 m tall



$$\sin 77 = \frac{x}{3.2}$$

$$x = 3.12 \text{ km}$$

$\therefore$ John is 3.12 km south of camp



$$\tan 55 = \frac{h}{y}$$

$$\tan 27 = \frac{h}{(425 + y)}$$

$$h = y \tan 55$$

$$h = (425 + y) \tan 27$$

$$\therefore y \tan 55 = (425 + y) \tan 27 \quad (\text{solve})$$

$$\therefore y = 235.73 \text{ m}$$

$$\therefore \cos 55 = \frac{235.73}{x}$$

$$\therefore EM \text{ is } \underline{\underline{410.99 \text{ m}}}$$

$$\therefore x = 410.99 \text{ m} \quad \checkmark$$

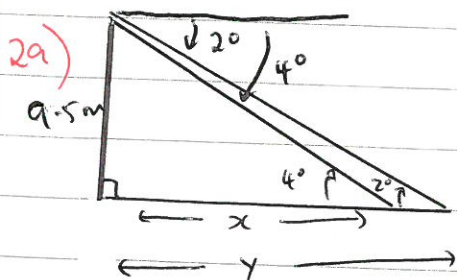
28b) $\cos 27 = \frac{(235.73 + 425)}{z}$

$$\therefore z = 741.56 \text{ m} \quad \checkmark$$

$$\underline{\underline{ER = 741.56 \text{ m}}}$$

28c) $h = 235.73 \tan 55$

$$\therefore \text{the height is } \underline{\underline{336.66 \text{ m}}}$$



$$\tan 2 = \frac{9.5}{y}$$

$$y = \underline{\underline{272.04 \text{ m}}}$$

$$\tan 4 = \frac{9.5}{x}$$

$$\therefore x = \underline{\underline{135.86 \text{ m}}}$$

$$\therefore \text{shark has swam } \underline{\underline{136.19 \text{ m}}}$$